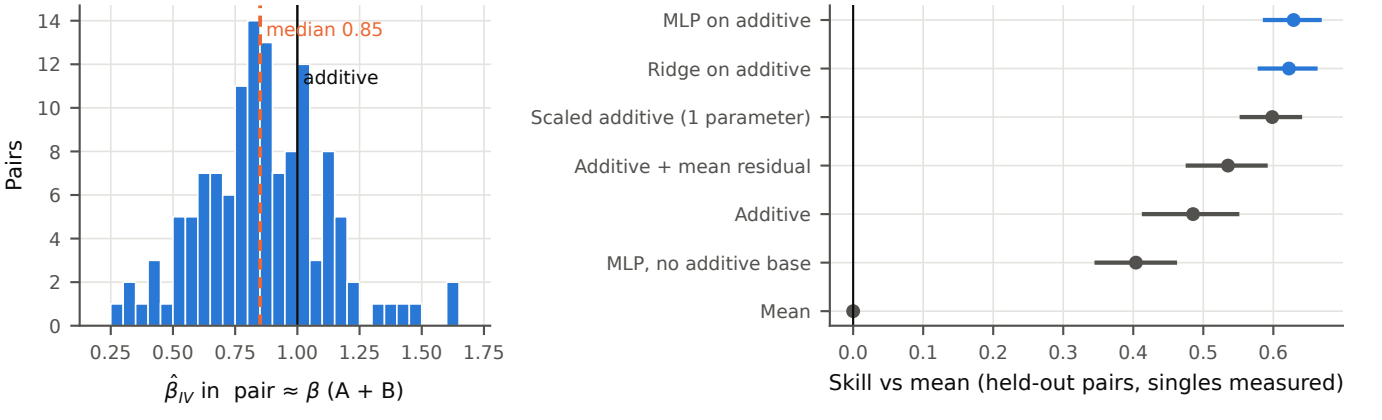
a Pairs respond less than the sum of singles
median 0.85
additive
Pairs
0
2
4
6
8
10
12
14
0.25
0.50
0.75
1.00
1.25
1.50
1.75
$\hat{\beta}_{IV}$ in pair ≈ β (A + B)
b Learned residuals add little to the scaled sum
MLP on additive
Ridge on additive
Scaled additive (1 parameter)
Additive + mean residual
Additive
MLP, no additive base
Mean
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Skill vs mean (held-out pairs, singles measured)